From Discrete Snapshots to Geometric Entity: Motion-Native Channel Tracking for THz Near-Field XL-MIMO

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***Abstract*—Channel tracking in mobile terahertz (THz) near-field extremely large-scale multiple-input multiple-output (XL-MIMO) systems is challenging because spherical-wave propagation couples angle and range, leading to rapid and nonlinear channel evolution under sparse pilot observations. Existing approaches typically estimate pilot-local channel or position snapshots before applying temporal tracking or smoothing, making them vulnerable to information loss during snapshot formation. To break this bottleneck, we propose a paradigm shift from tracking discrete snapshots to estimating a holistic geometric entity, namely, one shared continuous UE trajectory over the tracking window. Specifically, we parameterize the trajectory using a low-dimensional cubic B-spline representation and directly couple the per-pilot profile likelihoods through this shared trajectory before any pilot-local position estimate is finalized, thereby inducing continuous near-field channel evolution. We further characterize the efficient geometric information supplied by each pilot observation and its aggregation in the shared spline-coefficient space. Based on this formulation, we develop a regularized maximum a posteriori estimator and a quadratic-penalty lifted solver that separates parallel pilot-local geometric updates from a structured global spline update. The resulting global system is banded and, for a fixed spline degree, admits linear-complexity factorization, enabling an efficient structured implementation. Simulations demonstrate consistently improved tracking accuracy over representative temporally refined baselines in the tested medium-to-high-SNR regime, together with a favorable accuracy–runtime trade-off. These results highlight the benefit of coupling sparse pilot observations through a shared continuous trajectory rather than temporally processing independently estimated snapshots.**

***Index Terms*—Terahertz, XL-MIMO, Near-field, channel estimation, continuous-time channel tracking, B-spline modeling.**

I. Introduction

TERAHERTZ (THz) communications and extremely large-scale multiple-input multiple-output (XL-MIMO) are widely regarded as the twin pillars of high-capacity wireless systems for the upcoming 6G era [1]. By leveraging the enormous bandwidth at THz frequencies [2] and the high beamforming gain of XL-MIMO [3], these systems can theoretically achieve Tbps-level data rates. However, the deployment of physically large arrays introduces severe near-field propagation effects, where the wavefronts impinging on the array are spherical rather than planar [4]. Unlike conventional far-field channels that depend solely on angles, the near-field THz channel depends on both the user's angle and range. This added dimensionality makes instantaneous channel state information (CSI) sensitive to user movement and increases the demand for frequent pilot support in dynamic scenarios [5].

A. Related Work and Limitations

Existing near-field channel estimation methods can be broadly classified into sparsity-aware, parametric, and model-driven approaches [6]. Sparsity-aware methods exploit structured channel representations in an appropriate transform domain. Representative examples include on-grid polar-domain orthogonal matching pursuit (OMP) and off-grid P-SIGW [7], near-field sparse representation [8], subarray-based hybrid-field estimation [9], and Bayesian sparse reconstruction [10]–[12]. Parametric methods instead estimate a compact set of physical parameters directly in the continuous domain, including joint maximum-likelihood (ML) estimation [13], tensor-based methods [14], [15], and sensing- or motion-assisted estimation [13], [16], [17]. Model-driven methods further combine structured physical inference with trainable modules. Recent hybrid-field examples include learned-sparsifying-matrix Bayesian estimation, dual-attention deep-unfolded Bayesian learning, and model-driven deep learning [18]–[20].

These methods provide effective slot-wise channel or physical-parameter estimates and naturally serve as the front end of dynamic channel tracking. Existing tracking methods typically follow an estimate-then-track architecture. A static estimator first recovers channel states or physical parameters at individual pilot instants, after which temporal priors are used to propagate or smooth these estimates. Deterministic methods track physical parameters such as angle of arrival (AoA), delay, and Doppler through electromagnetic or kinematic models, using techniques such as extended Kalman filtering [21], [22]. Rauch–Tung–Striebel (RTS) smoothing [23] provides a state-space tracking reference. Statistical methods instead describe channel evolution through Markov-type models and track latent sparse states [24], [25]. Temporal consistency can also be introduced without a recursive state-space model. For example, penalized B-spline smoothing [26] fits a continuous trajectory to positions estimated independently at the pilot instants. Although this approach produces a continuous output, the pilot observations have already been compressed into separate snapshot estimates before the spline is fitted. The temporal

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TABLE I
POSITIONING OF REPRESENTATIVE NEAR-FIELD CHANNEL ESTIMATION AND TRACKING PARADIGMS.

Representative direction	**Inference target**	**Temporal coupling**	**Observation interface**	**Inference principle**
Polar-domain estimation [7]	Snapshot CSI	None	Pilot-local observations	Grid/off-grid sparse recovery
Parametric estimation [13]	Physical parameters	None	Pilot-local observations	Slot-wise likelihood fitting
Deterministic tracking [21], [23]	Discrete physical states	Kinematic evolution	Snapshot sequence	Estimate–then–track
Statistical tracking [24], [25]	Latent channel states	Statistical evolution	Snapshot sequence	Estimate–then–track
B-spline post-processing [26]	Fitted continuous trajectory	Spline smoothing	Position snapshots	Estimate–then–smooth
Proposed	**continuous channel**	**Window-level coupling**	**Joint likelihoods**	**Direct inference**

Note: Representative formulation-level characteristics; the table does not provide a numerical performance ranking.

model therefore refines the estimated snapshots rather than participating directly in the observation-level inference.

Despite their different estimators and temporal models, these approaches first compress each pilot observation into a local state during the estimation stage and impose temporal coupling afterward during the tracking stage. Under parameter-independent processing, observation-level information discarded during this compression cannot be recovered by subsequent temporal processing according to the data processing inequality [27], [28]. This motivates coupling the pilot observations directly through the shared trajectory itself.

It is also useful to distinguish the above tracking problem from sensing-assisted and integrated sensing and communication (ISAC) formulations [16], [21]. From the perspective relevant to this paper, ISAC primarily addresses how to enrich the available geometric observations through sensing and communication resources, whereas the present work addresses how sparse communication-pilot observations are jointly inferred over time. The two directions are therefore complementary.

B. Motivation and Contributions

Motivated by this distinction, we incorporate estimation and temporal tracking directly into the observation-level inference rather than imposing it on independently estimated snapshots. In this way, the continuous geometric process becomes the primary inference object, rather than a post-processed sequence of channel or position snapshots as illustrated in Fig. 1. The main contributions of this paper are summarized as follows:

- **Formulate a motion-native continuous channel tracking problem:** We reformulate sparse-pilot near-field channel tracking as direct inference of a shared continuous UE trajectory rather than independent pilot-local snapshots. A cubic B-spline representation is introduced to parameterize the trajectory, which couples all pilot observations through a common geometric entity before any pilot-local state is finalized. Unlike estimate-then-smooth pipelines, the temporal model acts on the per-pilot profile likelihoods before any position estimate is finalized, retaining the shared trajectory as the primary inference variable.
- **Characterize the information properties and derive the lifted optimization structure:** After analytically eliminating the array-common complex coefficients, we characterize the observation-supported identifiability conditions of the shared spline trajectory and quantify the information loss caused by snapshot-first processing. Based on these properties, we further develop a finite quadratic-penalty lifted formulation.
- **Develop a structure-exploiting local–global solver**: By leveraging the parallel separability of pilot-local updates in the lifted formulation and the compact support of B-spline bases, the proposed solver enables parallel local refinement and banded global updates, achieving linear-complexity factorization with respect to the number of spline coefficients for a fixed spline degree.

Extensive simulations further validate the resulting tracking gain, operating scope, optimization behavior, and tested accuracy–runtime trade-off.

C. Organization and Notation

The remainder of this paper is organized as follows. Section II introduces the system model and formulates the continuous channel-tracking problem. Section III develops the regularized MAP formulation, analyzes observation-supported identifiability, and introduces the lifted penalty surrogate. Section IV presents the alternating solver and its structured implementation. Section V reports the simulation results, and Section VI concludes the paper.

Notation: In this paper, bold uppercase and lowercase letters stand for matrices and vectors, respectively. Unless otherwise stated, all vectors are column vectors. Superscripts $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$ denote the transpose, conjugate transpose, and inverse operations. The vectorization of a matrix is denoted as $\mathrm{vec}(\cdot)$, whose reverse process is $\mathrm{devec}(\cdot)$. The estimation of a variable $\mathbf{X}$ is denoted as $\hat{\mathbf{X}}$. The symbol $\otimes$ stands for the Kronecker product and $\mathbf{I}_x$ is the $x \times x$ identity matrix.

II. SYSTEM MODEL

In this system, we assume different user-equipment (UE) terminals are assigned orthogonal pilots across the N-slot time resource block. Here, we consider a synchronized and calibrated system with dominant trackable LoS path and a fixed receiver-known combiner $\mathbf{W}$. Accordingly, we focus on a single typical UE served by a multi-antenna base station (BS) equipped with M antennas connected to N_{RF} radio-frequency (RF) chains. The horizontal axis of the array is aligned with global x-axis in the two-dimensional Cartesian coordinate system and we assume that the UE lies in the known service half-plane to remove the standard mirror ambiguity of a ULA in the following sections.

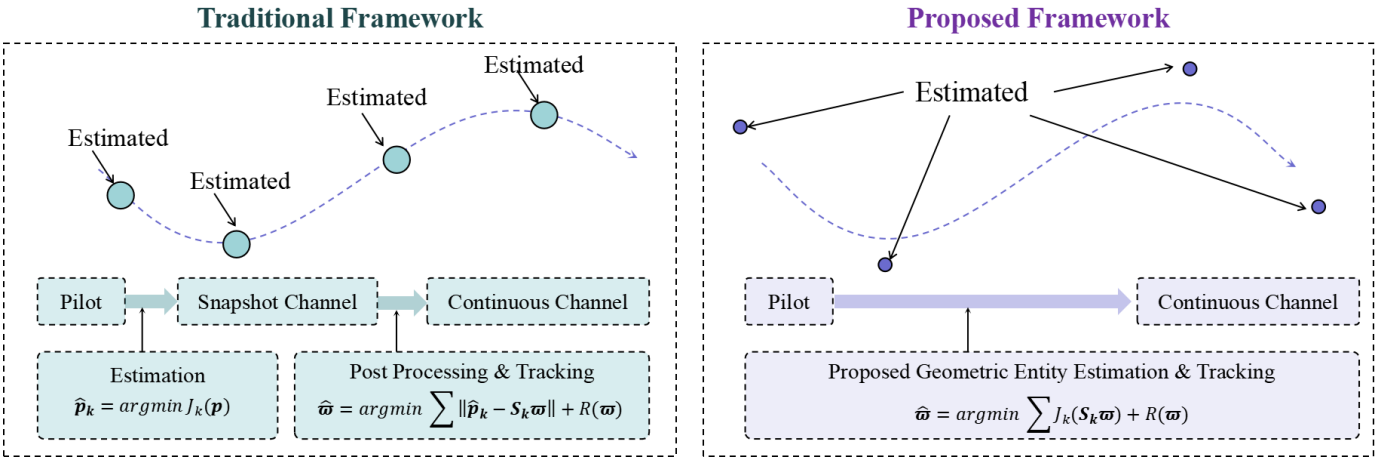

Fig. 1. From snapshot-first estimation and temporal post-processing to motion-native continuous near-field channel tracking.

In practical communication systems, the full observation window can be discretized as $\mathcal{T} = \{t_1, t_2, \cdots, t_N\}$ and observations are only available at a subset of instants, denoted by $\mathcal{O} = \{t_{n_1}, t_{n_2}, \cdots, t_{n_K}\}$, $K \ll N$. The K pilot-bearing slots provide the observations used for tracking, whereas the remaining $N-K$ slots may carry payload data or control signaling. Hence, K/N denotes temporal pilot support rather than complete time–frequency overhead, as illustrated in Fig. 2. In this paper, the geometric entity refers to the complete UE trajectory over one tracking window. Motion-native means that this trajectory enters the joint pilot likelihood before any pilot-local channel or position estimate is finalized.

A. THz Near-Field XL-MIMO Signaling Model

For a half-wavelength-spaced ULA, the position of mth antenna element is denoted as $\mathbf{s}_m = \left[(m-(M+1)/2)\,\frac{\lambda}{2}, 0\right]^{\mathsf{T}}$, where λ is the wavelength at carrier frequency f_c. The UE position at the kth pilot instant is denoted by $\mathbf{p}_k = \mathbf{p}(t_{n_k}) = [p_{x,k}, p_{y,k}]^{\mathsf{T}}$. Therefore, the propagation vector to the mth antenna element is $\mathbf{r}_{m,k} = \mathbf{p}_k - \mathbf{s}_m$ and the distance $d_m(\mathbf{p}) = |\mathbf{p} - \mathbf{s}_m|_2$. Based on these element-dependent propagation distances, the spherical-wave spatial response $\mathbf{a}(\mathbf{p}) \in \mathbb{C}^M$ in the near-field region is modeled as

$$[\mathbf{a}(\mathbf{p})]_m = \exp\left(-\mathrm{j}\frac{2\pi f_c}{c} d_m(\mathbf{p})\right). \tag{1}$$

Besides this position-dependent phase-only near field response, $\alpha(t)$ is an array-common complex coefficient that absorbs other slowly varying effects, including residual common phase, pathloss, and synchronization/calibration errors. Because $\alpha(t)$ usually varies much more slowly than the geometry-induced spatial response $\mathbf{a}(\mathbf{p})$, in this paper we assume that the coherence time of $\alpha(t)$ exceeds the inter-pilot interval. Accordingly, we adopt the blockwise-constant model $\alpha(t) = \alpha_k$ for $t \in [t_{n_k}, t_{n_{k+1}})$, $k = 1, \ldots, K-1$, with α_K used over the final interval $[t_{n_K}, t_N]$.

Thus, the dominant-path channel is modeled as $\mathbf{h}(t) = \alpha(t)\mathbf{a}(\mathbf{p}(t))\,, h_k = \mathbf{h}(t_{n_k})$ and the corresponding post-combining observations can be written as $\mathbf{Y}_k = \mathbf{W}\mathbf{h}_k\boldsymbol{\gamma}_k^{\mathsf{T}} + \mathbf{WN}_k \in \mathbb{C}^{N_{\mathrm{RF}} \times T_{\mathrm{P}}}$, where $\boldsymbol{\gamma}_k$ is the pilot block and $\mathbf{N}_k$ denotes the antenna-domain noise matrix. Equivalently, vectorizing the received $\mathbf{Y}_k$ gives

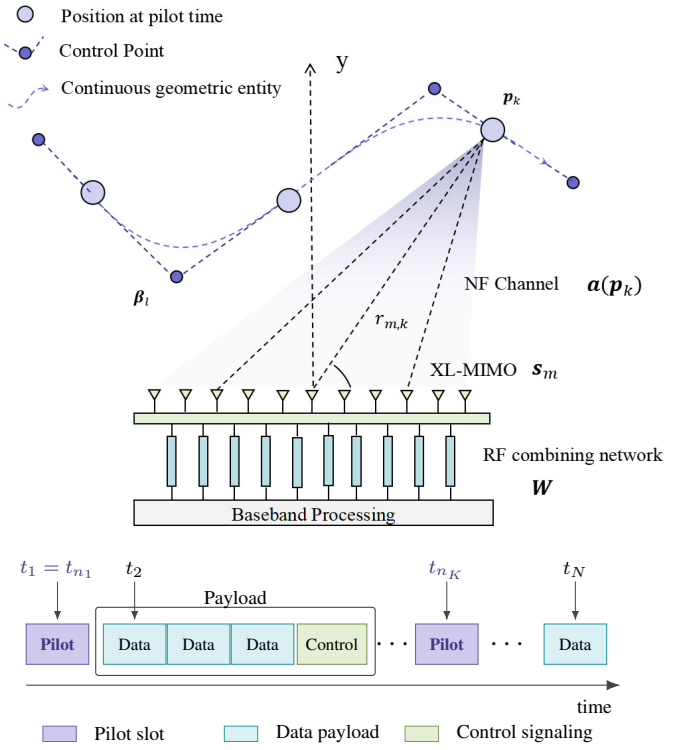

Fig. 2. Pilot-supported time grid with pilot-bearing and pilot-free physical slots.

$$\mathbf{y}_k = \alpha_k \boldsymbol{\psi}_k(\mathbf{p}_k) + \mathbf{n}_k, \tag{2}$$

where N_y denotes the dimension of the vectorized pilot observation, $\mathbf{y}_k, \mathbf{n}_k, \boldsymbol{\psi}_k(\mathbf{p}_k) \in \mathbb{C}^{N_y}$. $\mathbf{y}_k$ is the received signal, $\boldsymbol{\psi}_k(\mathbf{p}) \triangleq \boldsymbol{\gamma}_k \otimes \mathbf{W}\mathbf{a}(\mathbf{p})$ maps a trial UE position directly to the corresponding noiseless post-combining pilot response, α_k accounts for the common complex scaling. Since the combining matrix $\mathbf{W} \in \mathbb{C}^{N_{\mathrm{RF}} \times M}$ is fixed across the tracking window,

TABLE II
PRINCIPAL NOTATION AND DIMENSIONS.

Symbol and domain	Meaning
$\mathbf{p}_k(\boldsymbol{\varpi}) = \mathbf{S}_k\boldsymbol{\varpi} \in \mathbb{R}^2$	Trajectory position at t_{n_k}
$\mathbf{a}(\mathbf{p}) \in \mathbb{C}^M$	Near-field channel response
$\mathbf{W} \in \mathbb{C}^{N_{\mathrm{RF}} \times M}$	Fixed semiunitary combiner
$\boldsymbol{\psi}_k(\mathbf{p}) \in \mathbb{C}^{N_{\mathrm{y}}}$	Parameterized sensing vector
$\alpha_k \in \mathbb{C}$	Array-common nuisance coefficient
$\boldsymbol{\varpi} \in \mathbb{R}^{2L}$	Stacked B-spline coefficients
$\mathbf{S}_k \in \mathbb{R}^{2\times 2L}$	Spline sampling matrix at t_{n_k}
$\mathbf{z}_k \in \mathbb{R}^2$	Auxiliary local position variable
$\mathbf{D}_x \in \mathbb{R}^{2(L-x)\times 2L},\ x \in \{1,2\}$	x-order difference operator
$\mathbf{R} \in \mathbb{R}^{2L\times 2L}$	Total regularization matrix
$\mathbf{C}_{\boldsymbol{\varpi}} \in \mathbb{R}^{2L\times 2L}$	Global normal matrix
$\boldsymbol{\Gamma}_{\boldsymbol{\varpi}} \in \mathbb{R}^{2L}$	Global right-hand side

known at the receiver, and semiunitary $\mathbf{n}_k = vec(\mathbf{W}\mathbf{N}_k) \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_{N_y})$. Consequently, the corresponding likelihood can be expressed using Euclidean residuals with the common scaling factor $1/\sigma^2$.

B. Continuous-Time B-Spline Parametrization

Rather than treating positions as independent variables, we regard the continuous UE trajectory as a motion-native geometric entity. We represent the continuous UE trajectory by a degree-q B-spline with L two-dimensional control points $\boldsymbol{\beta}_\ell = [\beta_{x,\ell}, \beta_{y,\ell}]^{\mathsf{T}} \in \mathbb{R}^2$, stacked as

$$\boldsymbol{\varpi} \triangleq \left[\boldsymbol{\beta}_0^{\mathsf{T}}, \ldots, \boldsymbol{\beta}_{L-1}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{2L}, \tag{3}$$

where $L \ll N$ controls the dimension of the trajectory representation. Using normalized time $s(t) \triangleq (t-t_1)/(t_N-t_1) \in [0,1]$ and adopting a clamped open-uniform knot sequence, $\{\tau_i\}_{i=0}^{L+q}$ can be given by

$$\tau_i = \begin{cases} 0, & 0 \le i \le q, \\ \dfrac{i-q}{L-q}, & q+1 \le i \le L-1, \\ 1, & L \le i \le L+q, \end{cases} \tag{4}$$

The repeated endpoint knots clamp the trajectory at both ends. The basis functions follow the Cox–de Boor recursion [29], [30] as

$$B_{\ell,q}(s) = \frac{s-\tau_\ell}{\tau_{\ell+q}-\tau_\ell} B_{\ell,q-1}(s) + \frac{\tau_{\ell+q+1}-s}{\tau_{\ell+q+1}-\tau_{\ell+1}} B_{\ell+1,q-1}(s). \tag{5}$$

At degree zero, $B_{\ell,0}(s)$ equals one over its associated half-open knot interval and zero elsewhere, with the final interval including the right endpoint.

Collecting the L basis functions evaluated at $s(t)$, we define $\mathbf{b}(t) \triangleq \left[B_{0,q}(s(t)), \ldots, B_{L-1,q}(s(t))\right]^{\mathsf{T}} \in \mathbb{R}^L$ and the corresponding position basis matrix $\mathbf{S}(t) \triangleq \mathbf{b}^{\mathsf{T}}(t) \otimes \mathbf{I}_2 \in \mathbb{R}^{2\times 2L}$. The continuous trajectory generated by the control points can therefore be written equivalently as

$$\mathbf{p}(t;\boldsymbol{\varpi}) = \sum_{\ell=0}^{L-1} B_{\ell,q}(s(t))\boldsymbol{\beta}_\ell = \mathbf{S}(t)\boldsymbol{\varpi}. \tag{6}$$

At the kth pilot instant, we further define the sampling matrix $\mathbf{S}_k \triangleq \mathbf{S}(t_{n_k})$, thus $\mathbf{p}_k(\boldsymbol{\varpi}) = \mathbf{S}_k\boldsymbol{\varpi}$. Substituting $\mathbf{p}_k(\boldsymbol{\varpi})$ into the signaling model makes every pilot observation depend on the same global coefficient vector $\boldsymbol{\varpi}$. The B-spline representation therefore reduces the trajectory description from $2N$ grid-wise position variables to $2L$ coefficients, while introducing temporal coupling directly at the observation level throughout the tracking window.

We use cubic splines ($q = 3$), the lowest degree providing C^2-continuous trajectories. Their compact support activates at most four consecutive basis functions, preserving the banded global structure while providing sufficient smoothness for the considered motion.

C. Problem Formulation

Combining the near-field observation model with the spline parameterization, each pilot observation depends on the same global trajectory coefficient vector $\boldsymbol{\varpi}$ through $\mathbf{p}_k(\boldsymbol{\varpi}) = \mathbf{S}_k\boldsymbol{\varpi}$, while retaining its own array-common nuisance coefficient α_k. Our objective is to estimate $\boldsymbol{\varpi}$ and thereby recover the continuous trajectory $\mathbf{p}(t)$. For compactness, collect the nuisance coefficients as $\boldsymbol{\alpha} \triangleq [\alpha_1, \ldots, \alpha_K]^{\mathsf{T}} \in \mathbb{C}^K$. Under the white Gaussian post-combining noise model, the negative log-likelihood, up to an additive constant, is

$$J_0(\boldsymbol{\varpi}, \boldsymbol{\alpha}) = \frac{1}{\sigma^2} \sum_{k=1}^{K} \left\| \mathbf{y}_k - \alpha_k \boldsymbol{\psi}_k\big(\mathbf{S}_k\boldsymbol{\varpi}\big) \right\|_2^2. \tag{7}$$

The corresponding unregularized maximum-likelihood estimator is therefore

$$\mathrm{P}_0: \quad \left\{\widehat{\boldsymbol{\varpi}}, \widehat{\boldsymbol{\alpha}}\right\} = \arg\min_{\boldsymbol{\varpi}, \boldsymbol{\alpha}} J_0(\boldsymbol{\varpi}, \boldsymbol{\alpha}). \tag{8}$$

Although the spline parameterization reduces the trajectory representation complexity from $2N$ grid-wise position variables to $2L$ global coefficients, P_0 remains a coupled nonlinear estimation problem.

III. Spline-Based MAP Framework with a Lifted Penalty Surrogate

In this section, we incorporate regularization to obtain a stable MAP formulation and eliminate the array-common nuisance coefficients analytically through variable projection starting from P_0. We then discuss the local identifiability provided by the near-field observations and the complementary stabilizing role of regularization. Finally, we introduce a lifted optimization framework, separating nonlinear pilot-local fitting from global spline consistency.

A. Regularized MAP and Variable Projection

Because practical UE trajectories are typically smooth, we exploit temporal smoothness to stabilize P_0 in the sparse-pilot regime. From a Bayesian perspective, this regularization corresponds to an intrinsic Gaussian prior

$$p(\boldsymbol{\varpi}) \propto \exp[-\mathcal{R}(\boldsymbol{\varpi})], \tag{9}$$

which favors slowly varying control-point sequences through first- and second-order differences that penalize velocity- and

acceleration-like variations, respectively. In addition, since $\boldsymbol{\varpi}$ is the primary variable in this paper, the smoothness prior is imposed directly on the spline coefficients rather than on independently sampled positions.

Define the corresponding scalar difference operators for a generic coefficient sequence $\mathbf{x} = [x_0, \ldots, x_{L-1}]^{\mathsf{T}}$ as $\boldsymbol{\Delta}_1 \in \mathbb{R}^{(L-1)\times L}$ and $\boldsymbol{\Delta}_2 \in \mathbb{R}^{(L-2)\times L}$, where

$$[\boldsymbol{\Delta}_1\mathbf{x}]_\ell = x_{\ell+1} - x_\ell, \qquad [\boldsymbol{\Delta}_2\mathbf{x}]_\ell = x_{\ell+2} - 2x_{\ell+1} + x_\ell. \tag{10}$$

These scalar operators are then extended to the two-dimensional coefficient vector as

$$\mathbf{D}_x \triangleq \boldsymbol{\Delta}_x \otimes \mathbf{I}_2 \in \mathbb{R}^{2(L-x)\times 2L}, \qquad x \in \{1, 2\}. \tag{11}$$

Accordingly, $\mathbf{D}_1\boldsymbol{\varpi}$ captures the first-order variation between adjacent control points, whereas $\mathbf{D}_2\boldsymbol{\varpi}$ captures the variation of their successive increments. Therefore, the prior regularizer is $\mathcal{R}(\boldsymbol{\varpi}) = \frac{1}{2}\boldsymbol{\varpi}^{\mathsf{T}}\mathbf{R}\boldsymbol{\varpi}$ where

$$\mathbf{R} \triangleq \lambda_1\mathbf{D}_1^{\mathsf{T}}\mathbf{D}_1 + \lambda_2\mathbf{D}_2^{\mathsf{T}}\mathbf{D}_2 \in \mathbb{R}^{2L\times 2L}, \tag{12}$$

where nonnegative weights λ_1 and λ_2 control the penalties on control variation. Their effective values depend on the coordinate units and the relative scaling of the likelihood and regularization terms, and are therefore treated as tuning parameters. Combining this regularization with the data fidelity in (7) gives

$$J_{\mathrm{MAP}}(\boldsymbol{\varpi}, \boldsymbol{\alpha}) = J_0(\boldsymbol{\varpi}, \boldsymbol{\alpha}) + \mathcal{R}(\boldsymbol{\varpi}), \tag{13}$$

and the corresponding regularized MAP problem is

$$\mathrm{P}_1: \quad \{\widehat{\boldsymbol{\varpi}}, \widehat{\boldsymbol{\alpha}}\} = \arg\min_{\boldsymbol{\varpi}, \boldsymbol{\alpha}} J_{\mathrm{MAP}}(\boldsymbol{\varpi}, \boldsymbol{\alpha}). \tag{14}$$

Although the trajectory regularization stabilizes P_1, the global spline coefficients and the K array-common nuisance coefficients remain coupled through the nonlinear position-dependent sensing model. For a fixed trajectory coefficient vector $\boldsymbol{\varpi}$, the nuisance coefficient α_k enters the kth observation model linearly through the induced pilot position $\mathbf{p}_k = \mathbf{S}_k\boldsymbol{\varpi}$. For any trial position $\mathbf{p}$, α_k can be eliminated analytically by variable projection as

$$\alpha_k^\star(\boldsymbol{\varpi}) = \frac{\boldsymbol{\psi}_k^{\mathsf{H}}(\mathbf{S}_k\boldsymbol{\varpi})\mathbf{y}_k}{\boldsymbol{\psi}_k^{\mathsf{H}}(\mathbf{S}_k\boldsymbol{\varpi})\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi})}, \tag{15}$$

provided that $\boldsymbol{\psi}_k^{\mathsf{H}}(\mathbf{S}_k\boldsymbol{\varpi})\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi}) > 0$. After estimating $\widehat{\boldsymbol{\varpi}}$, the nuisance coefficients are recovered as $\widehat{\alpha}_k = \alpha_k^\star(\widehat{\boldsymbol{\varpi}})$, and the continuous channel is reconstructed accordingly. Substituting this closed-form solution into the observation residual yields

$$J_k(\mathbf{S}_k\boldsymbol{\varpi}) \triangleq \frac{1}{\sigma^2}\left\|\mathbf{y}_k - \alpha_k^\star(\mathbf{S}_k\boldsymbol{\varpi})\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi})\right\|_2^2. \tag{16}$$

where $J_k(\mathbf{p})$ measures the variance-normalized residual energy after the best array-common complex scaling has been removed at the trial position $\mathbf{p}$.

Applying this concentration step at all pilot instants converts P_1 into the equivalent reduced objective

$$J_{\mathrm{dir}}(\boldsymbol{\varpi}) = \sum_{k=1}^{K} J_k(\mathbf{S}_k\boldsymbol{\varpi}) + \mathcal{R}(\boldsymbol{\varpi}), \tag{17}$$

which leaves the global spline coefficient vector $\boldsymbol{\varpi}$ as the primary inference variable.

Variable projection leaves $\boldsymbol{\varpi}$ as the sole inference variable, but J_{dir} remains nonlinear because all pilots are coupled through the shared spline. We next characterize the observation-supported information of this representation before developing the solver.

B. Observation-Level Information and Identifiability

The shared trajectory representation reduces the unknown trajectory from discrete position samples to a finite-dimensional coefficient vector. However, this reduction alone does not guarantee that the coefficient vector is supported by the available pilot observations. Therefore, we characterize the efficient information of $\boldsymbol{\varpi}$ after eliminating the array-common nuisance coefficients, and distinguish observation-supported local identifiability from regularization-induced curvature. For the theoretical analysis below, all information quantities are evaluated at the true parameter values.

1) Identifiability of the Shared Trajectory Representation: For the kth pilot, the coefficient-space Jacobian for post-combining response vector is $\mathbf{Q}_{\boldsymbol{\varpi},k} \triangleq \nabla\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi})$. With a coefficient perturbation $\delta\boldsymbol{\varpi}$, only the component of this response variation orthogonal to the nuisance direction induced by the unknown coefficient α_k carries efficient information about $\boldsymbol{\varpi}$. Thus, we project out this nuisance direction using

$$\mathbf{P}_{\psi,k}^{\perp} \triangleq \mathbf{I}_{N_{\mathrm{y}}} - \frac{\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi})(\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi}))^{\mathsf{H}}}{\|\boldsymbol{\psi}_k(\mathbf{S}_k\boldsymbol{\varpi})\|_2^2}. \tag{18}$$

The efficient information contributed directly to the global coefficient vector by the kth raw pilot observation is

$$\mathcal{I}_{\boldsymbol{\varpi},k}^{\mathrm{obs}} \triangleq \frac{2|\alpha_k|^2}{\sigma^2}\,\mathrm{Re}\left\{\mathbf{Q}_{\boldsymbol{\varpi},k}^{\mathsf{H}}\mathbf{P}_{\psi,k}^{\perp}\mathbf{Q}_{\boldsymbol{\varpi},k}\right\}. \tag{19}$$

To expose how pilot-local geometric sensitivity is mapped into the shared coefficient space, we obtain $\mathbf{Q}_{\mathbf{p},k} \triangleq \nabla\boldsymbol{\psi}_k(\mathbf{p})$ by the chain rule $\mathbf{Q}_{\boldsymbol{\varpi},k} = \mathbf{Q}_{\mathbf{p},k}\mathbf{S}_k$. Thus, $\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}$ captures the independent pilot-local geometric information

$$\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} \triangleq \frac{2|\alpha_k|^2}{\sigma^2}\,\mathrm{Re}\left\{\mathbf{Q}_{\mathbf{p},k}^{\mathsf{H}}\mathbf{P}_{\psi,k}^{\perp}\mathbf{Q}_{\mathbf{p},k}\right\}. \tag{20}$$

while $\mathbf{S}_k^{\mathsf{T}}\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}\mathbf{S}_k$ maps it into the coefficient space, allowing different pilots to jointly constrain the same coefficients.

More specifically, under conditionally independent pilot observations, these coefficient-space contributions add as

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \triangleq \sum_{k=1}^{K}\mathcal{I}_{\boldsymbol{\varpi},k}^{\mathrm{obs}}. \tag{21}$$

The shared coefficient vector $\boldsymbol{\varpi}$ is first-order locally identifiable from the raw pilot observations if and only if $\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \succ \mathbf{0}$. To expose how the pilot-local information jointly constrains the shared coefficients, we define the sampling matrix $\mathbf{A}_{\mathcal{O}} \triangleq \left[(\sqrt{\mathcal{I}_{\mathbf{p},1}^{\mathrm{obs}}}\mathbf{S}_1)^{\mathsf{T}}, \ldots, (\sqrt{\mathcal{I}_{\mathbf{p},K}^{\mathrm{obs}}}\mathbf{S}_K)^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{2K\times 2L}$, where $\sqrt{\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}}$ denotes its symmetric positive-semidefinite square root. For the adopted B-spline representation, further define $\mathbf{B}_{\mathcal{O}} \triangleq [\mathbf{b}(t_{n_1}), \ldots, \mathbf{b}(t_{n_K})]^{\mathsf{T}} \in \mathbb{R}^{K\times L}$. Therefore, this yields the following proposition.

Proposition 1 (Information-weighted spline identifiability): The shared coefficient vector $\boldsymbol{\varpi}$ is observation-supported first-order locally identifiable according to

$$\begin{aligned}\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \succ \mathbf{0} &\Longleftrightarrow \operatorname{rank}(\mathbf{A}_{\mathcal{O}}) = 2L \\ &\Longrightarrow \sum_{k=1}^{K} \operatorname{rank}\big(\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}\big) \geq 2L \Longrightarrow K \geq L,\end{aligned} \tag{22}$$

whereas, if $\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} \succ \mathbf{0}$ for all k, the exact condition reduces to

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \succ \mathbf{0} \quad \Longleftrightarrow \quad \operatorname{rank}(\mathbf{B}_{\mathcal{O}}) = L. \tag{23}$$

The proofs are given in Appendix A. Condition (22) allows rank-deficient pilot-local information as long as the mapped directions in $\mathbf{S}_k$ are complementary. Hence, a $2L$-dimensional spline trajectory can remain identifiable for $L \leq K \leq N$, whereas independently parameterized position cannot. Under full-rank local information, identifiability further requires $\operatorname{rank}(\mathbf{B}_{\mathcal{O}}) = L$, so $K \geq L$ alone is insufficient unless the pilot instants cover all spline coefficient directions.

These rank conditions concern observation-supported identifiability, whereas regularization contributes additional prior-induced curvature:

$$\mathbf{G}_{\boldsymbol{\varpi}}^{\mathrm{pen}} = \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} + \mathbf{R}. \tag{24}$$

Regularization does not add observation information but only supplies prior curvature along weakly constrained coefficient directions. Its role is to suppress excessive oscillations of the spline trajectory under sparse and noisy observations, rather than to provide additional observation information.

2) Information Preservation under Snapshot Compression: Proposition 1 shows how pilot-local geometric information can become complementary after being mapped into the shared spline-coefficient space. A remaining question is whether the same benefit can be retained by first compressing each pilot observation into an independent snapshot and imposing the trajectory model afterward.

Consider the parameter-independent snapshot-first processing chain

$$\mathbf{y} \xrightarrow{\mathcal{K}_{\mathrm{s}}} \widehat{\boldsymbol{\xi}} \xrightarrow{\mathcal{K}_{\mathrm{p}}} \widehat{\boldsymbol{\zeta}}, \tag{25}$$

where $\widehat{\boldsymbol{\xi}}$ denotes the pilot-local snapshot sequence and $\widehat{\boldsymbol{\zeta}}$ its temporally processed output. Let $\mathcal{I}_{\mathbf{p},k}^{\mathrm{snap}}$ denote the pilot-local efficient information retained after snapshot formation, and define the information losses

$$\Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}} \triangleq \mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} - \mathcal{I}_{\mathbf{p},k}^{\mathrm{snap}}, \qquad \Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} \triangleq \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{post}}, \tag{26}$$

together with $\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} \triangleq \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{post}}$. The resulting loss satisfies the following proposition.

Proposition 2 (Irrecoverable information loss under snapshot-first processing): For the adopted shared B-spline representation, the total trajectory-information loss under snapshot-first processing satisfies

$$\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} = \sum_{k=1}^{K} \mathbf{S}_k^{\mathsf{T}} \Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}} \mathbf{S}_k + \Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} \succeq \mathbf{0}. \tag{27}$$

Furthermore,

$$\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} = \mathbf{0} \Longleftrightarrow \Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}} = \mathbf{0},\ \forall k, \quad \Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} = \mathbf{0}. \tag{28}$$

The proof is provided in Appendix B.

Together, Propositions 1 and 2 capture the two sides of the proposed motion-native inference. The shared spline aggregates complementary pilot information, whereas information lost before this aggregation cannot be recovered by subsequent temporal processing. This motivates retaining direct observation-level coupling in J_{dir} rather than adopting a snapshot-then-smooth pipeline.

C. Lifted Surrogate and Block Decomposition

To solve the nonlinear coupling in $J_{\mathrm{dir}}(\boldsymbol{\varpi})$, we introduce pilot-local auxiliary variables $\mathbf{z}_k \in \mathbb{R}^2$ and promote local–global consistency through the finite quadratic penalty $Z_k(\mathbf{z}_k, \boldsymbol{\varpi}) = \|\mathbf{z}_k - \mathbf{S}_k\boldsymbol{\varpi}\|_2^2$. The lifting introduces auxiliary optimization variables without changing the physical inference target $\boldsymbol{\varpi}$.

Consequently, the global continuous trajectory remains the primary inference object, while the auxiliary variables merely expose the pilot-wise geometry needed for efficient local updates. The resulting finite quadratic-penalty objective is

$$J_\eta(\boldsymbol{\varpi}, \{\mathbf{z}_k\}) = \sum_{k=1}^{K} \left[J_k(\mathbf{z}_k) + \frac{\eta}{2} \mathcal{Z}_k(\mathbf{z}_k, \boldsymbol{\varpi}) \right] + \mathcal{R}(\boldsymbol{\varpi}), \tag{29}$$

so that $\{\widehat{\boldsymbol{\varpi}}, \{\widehat{\mathbf{z}}_k\}_{k=1}^K\} = \arg\min J_\eta(\boldsymbol{\varpi}, \{\mathbf{z}_k\})$. The penalty encourages but does not impose exact equality for finite η, which naturally leads to a block-coordinate strategy. By alternating between the $\{\mathbf{z}_k\}_{k=1}^K$ and the global spline coefficients $\boldsymbol{\varpi}$, the problem decomposes into K independent local-position updates and one global spline-coefficient update, as illustrated in Fig 3. For fixed $\boldsymbol{\varpi}^{(i)}$, the local subproblems are

$$\mathrm{P}_{z_k}: \quad \mathbf{z}_k^* = \arg\min_{\mathbf{z}_k \in \mathbb{R}^2} J_k(\mathbf{z}_k) + \frac{\eta}{2} \mathcal{Z}_k(\mathbf{z}_k, \boldsymbol{\varpi}), \tag{30}$$

Once the $\{\mathbf{z}_k^{(i+1)}\}_{k=1}^K = \{\mathbf{z}_k^*\}_{k=1}^K$ have been updated, they are fixed for the global spline-coefficient update:

$$\mathrm{P}_{\boldsymbol{\varpi}}: \ \boldsymbol{\varpi}^* = \arg\min_{\boldsymbol{\varpi}} \left[\mathcal{R}(\boldsymbol{\varpi}) + \frac{\eta}{2} \sum_{k=1}^{K} \mathcal{Z}_k(\mathbf{z}_k, \boldsymbol{\varpi}) \right]. \tag{31}$$

IV. Alternating Solver for the Lifted Motion-Native Penalty Problem

Building on the lifted objective and block subproblems in (29)–(31), this section specifies their numerical solution. We first present the local and global updates of one alternating cycle, then exploit their parallel and B-spline-induced structure for implementation and complexity analysis, and finally state the fixed-stage descent property.

A. Alternating Local and Global Updates

At alternating cycle i, the K local subproblems are solved in parallel with $\varpi^{(i)}$ fixed, using coarse search followed by safeguarded LM refinement.

For the coarse stage, let $\Omega \subset \mathbb{R}^2$ be a finite set of sampled offsets around the current spline position. The corresponding candidate set is

$$\mathcal{G}_k \triangleq \left\{ \mathbf{S}_k\boldsymbol{\varpi}^{(i)} + \Delta\mathbf{z} : \Delta\mathbf{z} \in \Omega \right\} \cup \left\{ \mathbf{z}_k^{(i)} \right\}, \tag{32}$$

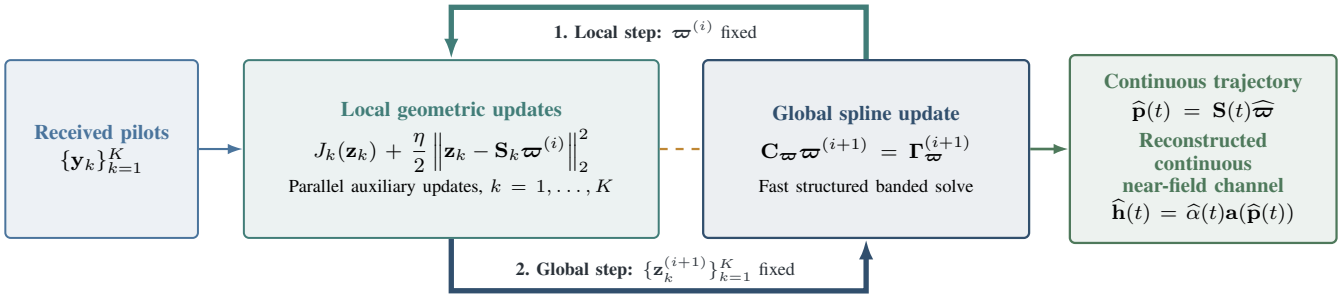

Fig. 3. Lifted local–global decomposition of the motion-native estimator.

whose first part samples nearby positions and second part retains the incumbent local variable. The coarse initializer is then selected by evaluating the actual local objective over this finite set to provide a reliable starting point for the subsequent safeguarded LM refinement:

$$\mathbf{z}_k^{(i,0)} = \arg\min_{\mathbf{z}\in\mathcal{G}_k}\left[J_k(\mathbf{z}) + \frac{\eta}{2}\left\|\mathbf{z} - \mathbf{S}_k\boldsymbol{\varpi}^{(i)}\right\|_2^2\right]. \tag{33}$$

Starting from $\mathbf{z}_k^{(i,0)}$, define the profiled residual and its position Jacobian as $\mathbf{r}_k(\mathbf{z}) \triangleq \mathbf{y}_k - \alpha_k(\mathbf{z})\boldsymbol{\psi}_k(\mathbf{z})$ and $\mathbf{J}_{r,k}(\mathbf{z}) \triangleq \nabla_{\mathbf{z}}\mathbf{r}_k(\mathbf{z})$. The gradient and Gauss–Newton curvature of the local objective are then modeled by

$$\begin{aligned}\mathbf{g}_k &= \frac{2}{\sigma^2}\,\mathrm{Re}\left\{\mathbf{J}_{r,k}^{\mathsf{H}}\mathbf{r}_k\right\} + \eta\left(\mathbf{z} - \mathbf{S}_k\boldsymbol{\varpi}^{(i)}\right),\\ \mathbf{H}_k &= \frac{2}{\sigma^2}\,\mathrm{Re}\left\{\mathbf{J}_{r,k}^{\mathsf{H}}\mathbf{J}_{r,k}\right\} + \eta\mathbf{I}_2.\end{aligned} \tag{34}$$

All position-dependent quantities in (34) are evaluated at the current local iterate. Given a damping parameter $\mu \geq 0$, the LM candidate and the safeguarded update are

$$\begin{aligned}\widetilde{\mathbf{z}}_k &= \mathbf{z} - \left(\mathbf{H}_k + \mu\mathbf{I}_2\right)^{-1}\mathbf{g}_k,\\ \mathbf{z}_k^{+} &\in \arg\min_{\mathbf{u}\in\{\mathbf{z},\widetilde{\mathbf{z}}_k\}}\left[J_k(\mathbf{u}) + \frac{\eta}{2}\mathcal{Z}_k(\mathbf{u},\boldsymbol{\varpi})\right].\end{aligned} \tag{35}$$

The second line evaluates the actual local objective and retains the current iterate whenever the LM candidate fails to decrease it. After local refinement terminates, we set $\mathbf{z}_k^{(i+1)} = \mathbf{z}_k^{+}$. The same procedure is carried out independently for all k, enabling parallel execution.

After the parallel local updates, fixing $\{\mathbf{z}_k^{(i+1)}\}_{k=1}^K$ yields a quadratic global subproblem in $\boldsymbol{\varpi}$. Setting the gradient of (31) with respect to $\boldsymbol{\varpi}$ to zero gives

$$\mathbf{C}_{\boldsymbol{\varpi}}\boldsymbol{\varpi}^{(i+1)} = \boldsymbol{\Gamma}_{\boldsymbol{\varpi}}^{(i+1)}, \tag{36}$$

where the corresponding global normal matrix $\mathbf{C}_{\boldsymbol{\varpi}} \in \mathbb{R}^{2L\times 2L}$ and right-hand side vector $\boldsymbol{\Gamma}_{\boldsymbol{\varpi}}^{(i+1)} \in \mathbb{R}^{2L}$ are specified as

$$\begin{aligned}\mathbf{C}_{\boldsymbol{\varpi}} &\triangleq \mathbf{R} + \eta\sum_{k=1}^{K}\mathbf{S}_k^{\mathsf{T}}\mathbf{S}_k,\\ \boldsymbol{\Gamma}_{\boldsymbol{\varpi}}^{(i+1)} &\triangleq \eta\sum_{k=1}^{K}\mathbf{S}_k^{\mathsf{T}}\mathbf{z}_k^{(i+1)} \in \mathbb{R}^{2L}.\end{aligned} \tag{37}$$

When $\mathbf{C}_{\boldsymbol{\varpi}} \succ \mathbf{0}$, (36) admits a unique global minimizer. Algorithm 1 summarizes the complete alternating solver. We next examine how the B-spline structure determines the fast implementation and reduces computational cost.

Algorithm 1 Alternating Solver for the Lifted Penalty Surrogate

Require: $\{\mathbf{y}_k, t_{n_k}, \gamma_k\}_{k=1}^K$; fixed $\mathbf{W}$; $\{\mathbf{S}_k\}$, $\mathbf{R}$, $\{\eta_s\}_{s=1}^S$; solver tolerances.
Ensure: $\widehat{\boldsymbol{\varpi}}$, $\widehat{\mathbf{p}}(t) = \mathbf{S}(t)\widehat{\boldsymbol{\varpi}}$, and the reconstructed continuous channel $\widehat{\mathbf{h}}(t)$.

```
Initialize ϖ and {z_k}.
for s = 1, ..., S do
    Set η ← η_s.
    for i = 0, ..., I_max,s − 1 do
        for k = 1, ..., K in parallel do
            Select the coarse candidate using (33).
            Refine using (34) and apply the safeguard (35).
            Set the accepted result as z_k^(i+1).
        end for
        Form C_ϖ and Γ_ϖ^(i+1), and solve (36).
        if the fixed-stage stopping tests pass then
            break.
        end if
    end for
end for
Reconstruct p̂(t) = S(t)ϖ̂.
Recover {α̂_k}_{k=1}^K and reconstruct ĥ(t).
return ĥ(t).
```

B. Structured Fast Implementation

We next exploit the local support of the B-spline basis and the shared Kronecker structure to characterize the resulting global sparsity. The required properties are established in the following two lemmas, with proofs provided in Appendix C.

Lemma 1 (Local support of the B-spline basis): For the degree-q B-spline basis associated with the knot $\{\tau_i\}$, its support satisfies $\mathrm{supp}(B_{\ell,q}) \subseteq [\tau_\ell, \tau_{\ell+q+1}]$, $\ell = 0, \ldots, L-1$. Hence, at most $q+1$ consecutive basis functions are nonzero.

Lemma 2 (Banded sparsity of the global normal matrix): The matrix $\mathbf{C}_{\boldsymbol{\varpi}}$ is block banded with block half-bandwidth $b_{\mathbf{C}} \triangleq \max\{q, 2\}$ for $\ell, r = 0, \ldots, L-1$, independently of L.

As Lemmas 1 and 2 reveal the banded structure of $\mathbf{C}_{\boldsymbol{\varpi}}$, we further exploit the common Kronecker structure already embedded in $\mathbf{S}_k$ and $\mathbf{D}_x$, which reduces the $2L$-dimensional global update to two identical L-dimensional systems. Specif-

TABLE III
COMPLEXITY BEFORE AND AFTER EXPLOITING THE LOCAL–GLOBAL STRUCTURE.

Computational component	Serial and dense reference	Parallel and banded implementation
Local block per cycle	$\mathcal{O}(KMN_{\mathrm{RF}})$	$\mathcal{O}(MN_{\mathrm{RF}})$
Global setup and one solve	$\mathcal{O}(KL^2+L^3)$	$\mathcal{O}\big(K(q+1)^2+Lb_{\mathbf{C}}^2\big)$
Repeated global update	$\mathcal{O}(KL+L^2)$	$\mathcal{O}(K(q+1)+Lb_{\mathbf{C}})$
Complete alternating cycle	$\mathcal{O}\big(KMN_{\mathrm{RF}}+KL+L^2\big)$	$\mathcal{O}(MN_{\mathrm{RF}}+K(q+1)+Lb_{\mathbf{C}})$

Note: For the local block, the parallel entry denotes ideal execution latency rather than total arithmetic work.

ically, we have $\mathbf{C}_{\boldsymbol{\varpi}} = \mathbf{C}_s(\eta)\otimes\mathbf{I}_2$, where

$$\mathbf{C}_s(\eta) \triangleq \eta\sum_{k=1}^{K}\mathbf{b}(t_{n_k})\mathbf{b}^{\mathsf{T}}(t_{n_k})+\lambda_1\boldsymbol{\Delta}_1^{\mathsf{T}}\boldsymbol{\Delta}_1+\lambda_2\boldsymbol{\Delta}_2^{\mathsf{T}}\boldsymbol{\Delta}_2. \quad (38)$$

Since each $\mathbf{b}(t_{n_k})$ has at most $q+1$ active entries and the difference operators are naturally banded, $\mathbf{C}_s(\eta)$ also has a fixed bandwidth independent of L. Therefore, the two trajectory coordinates share the same banded system matrix, so the global update requires no explicit matrix inverse.

Rather than explicitly forming $\mathbf{C}_{\varpi}^{-1}$, we factorize the scalar banded matrix as $\mathbf{C}_s(\eta)=\mathbf{L}_s\mathbf{L}_s^{\mathsf{T}}$, where $\mathbf{L}_s$ is the triangular banded Cholesky factor, and each global update is computed by two banded triangular solves without explicitly forming $\mathbf{C}_{\boldsymbol{\varpi}}^{-1}$:

$$\begin{aligned}(\mathbf{L}_s\otimes\mathbf{I}_2)\mathbf{v}^{(i+1)} &= \boldsymbol{\Gamma}_{\boldsymbol{\varpi}}^{(i+1)},\\ (\mathbf{L}_s^{\mathsf{T}}\otimes\mathbf{I}_2)\boldsymbol{\varpi}^{(i+1)} &= \mathbf{v}^{(i+1)},\end{aligned} \quad (39)$$

where $\mathbf{v}^{(i+1)}$ is only the intermediate result of the forward substitution. Moreover, $\mathbf{C}_s(\eta)$ remains fixed within each continuation stage, so its $\mathcal{O}(Lb_{\mathbf{C}}^2)$ banded Cholesky factorization is performed only once and reused across alternating cycles. For fixed spline degree, $b_{\mathbf{C}}$ is constant, making both the factorization and repeated solves linear in L.

C. Algorithmic Efficiency and Fixed-Stage Descent

We now quantify the ideal parallel local-block cost, the banded global-block cost, and the fixed-stage descent property.

1) Local-Block Complexity: With $\boldsymbol{\varpi}^{(i)}$ fixed, the K local subproblems are independent. Each refinement is dominated by evaluating the dense-combined near-field response and its position derivatives, requiring $\mathcal{O}(MN_{\mathrm{RF}})$ time. Serial execution therefore scales as $\mathcal{O}(KMN_{\mathrm{RF}})$, whereas concurrent execution reduces the ideal local-block time to

$$T_{\mathrm{loc}}^{(\mathrm{cycle})} = \mathcal{O}(MN_{\mathrm{RF}}). \quad (40)$$

Thus, parallel execution removes the factor K from the ideal local-block latency.

2) Global-Block and Complete-Cycle Complexity: Without exploiting the B-spline locality and Kronecker structure, dense assembly and factorization of the global system require $\mathcal{O}(KL^2+L^3)$ operations. In the structured implementation, computing the banded Cholesky factorization [31] requires $\mathcal{O}\big(K(q+1)^2+Lb_{\mathbf{C}}^2\big)$.

Within each continuation stage, $\mathbf{C}_s(\eta)$ remains fixed and is therefore factorized only once. Each subsequent global update requires $\mathcal{O}(K(q+1))$ operations for right-hand-side assembly and $\mathcal{O}(Lb_{\mathbf{C}})$ operations for the banded forward–backward substitutions. The banded representation also reduces the global-system storage from $\mathcal{O}(L^2)$ to $\mathcal{O}(Lb_{\mathbf{C}})$. Combining this global update with the ideal parallel local block gives

$$T^{(\mathrm{cycle})} = \mathcal{O}(MN_{\mathrm{RF}}+K(q+1)+Lb_{\mathbf{C}})\,. \quad (41)$$

Hence, parallelization removes the factor K from the ideal local-block latency, while the fixed-bandwidth triangular solves reduce the repeated global update from quadratic to linear scaling in L, as summarized in Table III.

Importantly, the efficiency gains summarized above do not modify the underlying block updates. Parallelization only changes the execution order of the mutually independent local subproblems, while the banded implementation solves the same global normal equations as dense linear algebra. The structured implementation therefore preserves the descent property of the original alternating scheme.

For fixed η, the safeguarded local updates are nonincreasing and the global step exactly minimizes the quadratic $\boldsymbol{\varpi}$-subproblem. Hence, each alternating cycle satisfies

$$\begin{aligned}J_\eta\Big(\boldsymbol{\varpi}^{(i+1)},\{\mathbf{z}_k^{(i+1)}\}\Big) &\le J_\eta\Big(\boldsymbol{\varpi}^{(i)},\{\mathbf{z}_k^{(i+1)}\}\Big)\\ &\le J_\eta\Big(\boldsymbol{\varpi}^{(i)},\{\mathbf{z}_k^{(i)}\}\Big)\,.\end{aligned} \quad (42)$$

Therefore, the structured parallel implementation preserves monotonic descent. Since J_η is bounded below, its objective-value sequence converges within each continuation stage. This provides a clear fixed-stage stability guarantee for the proposed alternating solver. However, objective-value convergence does not by itself establish convergence of the iterates to a stationary point or a global optimum of the underlying nonconvex problem. Establishing such stronger guarantees requires substantially deeper theoretical analysis and constitutes an important direction for future work.

V. SIMULATION AND VALIDATION

A. Simulation Setup and Comparison Overview

To ensure a fair comparison, all methods are evaluated under the same physical geometry, observation model, and evaluation protocol, so that the resulting performance differences reflect estimator and solver design.

1) Simulation Configuration: Unless otherwise stated, all experiments use the nominal settings in Table IV, with following study-specific variations reported in the corresponding subsections. Throughout the simulations, the reported SNR denotes the nominal pre-combining per-antenna pilot SNR.

TABLE IV
MAIN SIMULATION AND SOLVER SETTINGS.

Parameter	Symbol	Value
Carrier frequency	f_c	100 GHz
Number of antennas	M	256
Number of RF chains	N_{RF}	16 to 256(default)
Element spacing	d	1.5 mm ($=\lambda/2$)
Observation duration	T	20 s
Number of timeslots	N	200
Number of pilot slots	K	10
Pilot symbol	γ_k	$1+\mathrm{j}$
Default SNR	SNR	20dB
Spline degree	q	3
Control points Number	L	8
Knot sequence	$\{\tau_\ell\}$	Uniform
Regularization weights	λ_1, λ_2	$10^{-6}, 10^{-4}$
Continuation schedule	η_s	10^{-2}(default) to 20
Coarse-search offset set	Ω	5×5 grids over $[-10,10]^2$ m
Global initialization	$\boldsymbol{\varpi}^{(0)}$	Interpolated straight line
Maximum cycles	$I_{\max,s}$	16

The BS is located at $[0,0]^{\mathsf{T}}$, and the centered ULA is aligned with the global x-axis. The aperture is $D=(M-1)\lambda/2=382.5$ mm, yielding the Rayleigh distance

$$R_R=\frac{2D^2}{\lambda}\approx 97.5 \text{ m}. \tag{43}$$

The principal experiments satisfy $r_{\max}<R_R$, except for the distance sweep, which crosses R_R to evaluate tracking performance around the conventional near–far-field boundary.

2) Compared Baselines: To compare with conventional estimate-then-track pipelines, we consider three snapshot estimators and three temporal-processing strategies.

The snapshot-estimation stage includes:

- **On-grid OMP** [7], which performs polar-domain sparse recovery over a predefined dictionary;
- **Off-grid P-SIGW** [7], which refines the spatial parameters in the continuous domain to mitigate grid mismatch; and
- **Joint ML** [13], which directly estimates the continuous position at each pilot instant under the adopted observation model.

The temporal-processing stage includes:

- **Raw output**, which retains the independently estimated pilot positions;
- **LS B-spline smoothing** [26], which fits a continuous trajectory to the position snapshots; and
- **RTS smoothing** [23], which serves as a state-space smoothing reference.

The resulting baselines are OMP+Raw, P-SIGW+Raw, Joint-ML+Raw, OMP+RTS, OMP+LS, P-SIGW+LS, and Joint-ML+LS. These pipelines estimate discrete snapshots before imposing temporal consistency, whereas the proposed method directly couples all pilot observations.

To ensure a fair comparison, all temporally refined methods receive the same ground-truth start and end positions as common boundary side information, so the comparison focuses on interior trajectory reconstruction rather than boundary initialization. For the proposed method, these endpoints are imposed through the quadratic anchor

$$J_{\mathrm{anc}}(\boldsymbol{\varpi})=\frac{\kappa_{\mathrm{anc}}}{2}\sum_{r\in\{\mathrm{s,e}\}}\|\mathbf{S}(t_r)\boldsymbol{\varpi}-\mathbf{p}_r^{\star}\|_2^2, \tag{44}$$

where $t_{\mathrm{s}}=t_1$, $t_{\mathrm{e}}=t_N$, and $\kappa_{\mathrm{anc}}=10^6$. In the simulations, $J_{\mathrm{anc}}(\boldsymbol{\varpi})$ is added to the global spline subproblem in equation (31), equivalently, it contributes the corresponding fixed quadratic terms to $\mathbf{C}_{\varpi}$ and Γ_{ϖ}, without altering the local updates or the banded structure of the solver. LS baselines impose the same endpoints as equality constraints, whereas RTS uses them as high-confidence boundary updates.

3) Common Comparison and Evaluation Protocol: In the following experiments, $\mathbf{W}$ is constructed from the N_{RF} rows of a fixed QR-orthogonalized complex-Gaussian matrix, yielding nested dense semiunitary combiners shared by all methods. Unless otherwise stated, the reported results are executed over $N_{\mathrm{MC}}=60$ paired Monte Carlo trials, which reduces the influence of occasional large-error realizations.

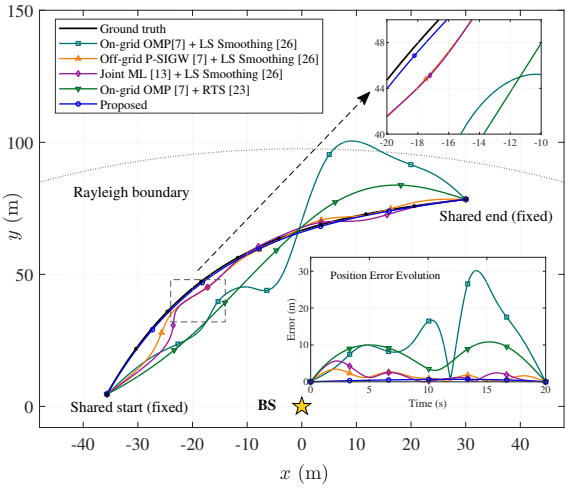

Fig. 4. Trajectory reconstruction and pointwise position error for one representative realization.

B. Channel Tracking Accuracy

We first evaluate the proposed framework through one representative trajectory realization and two statistical NMSE studies. Specifically, Fig. 4 presents a representative trajectory-level comparison and shows how different methods reconstruct the channel evolution between sparse pilot instants. Fig. 5 evaluates the NMSE over a broad SNR range, while Fig. 6 examines the NMSE at different propagation distances. Together, these experiments characterize both representative and statistical tracking performance.

As shown in Fig. 4, all temporally refined methods share the same endpoints under the common comparison protocol. The proposed method more accurately reconstructs the interior trajectory and maintains a consistently low position error, whereas the snapshot-first pipelines exhibit larger local deviations.

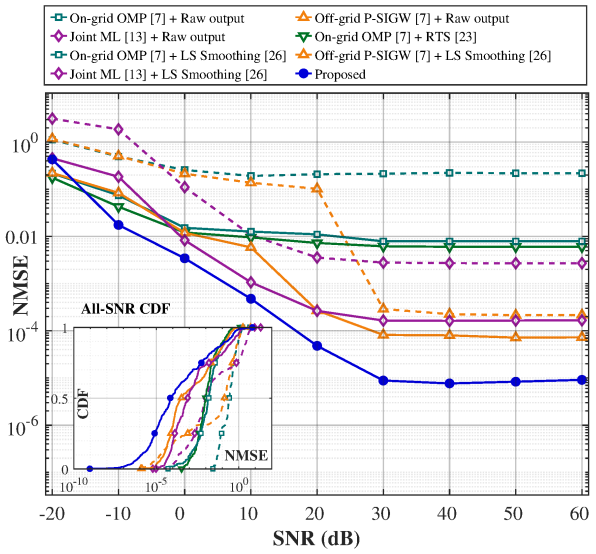

Fig. 5. NMSE versus SNR with the pooled empirical CDF over the tested SNR range.

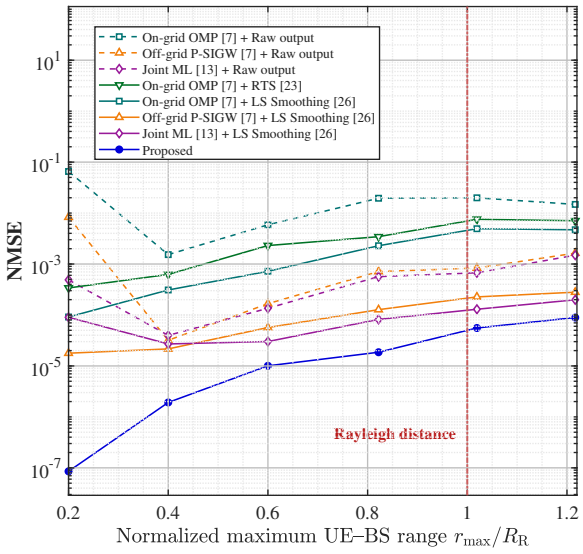

Fig. 6. NMSE versus normalized maximum UE–BS range.

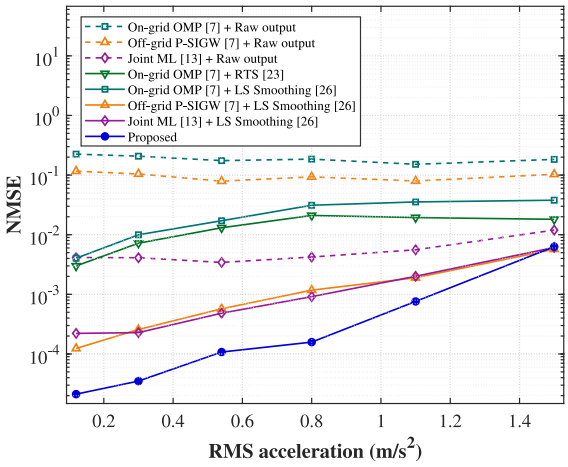

Fig. 7. NMSE versus RMS acceleration.

Fig. 5 further evaluates the channel-tracking performance over a broad SNR range. As the SNR increases, the proposed method continues to reduce the NMSE and achieves a clear advantage over the snapshot-first baselines. This trend is consistent with the information perspective in Sec. III-B. When the pilot observations become sufficiently informative, directly aggregating their geometric constraints through the shared trajectory avoids committing prematurely to independently estimated snapshots. In contrast, errors introduced during pilot-local estimation and subsequent temporal processing remain in the snapshot-first pipelines. The pooled empirical CDF over the tested SNR range further shows that the proposed method is consistently concentrated toward lower NMSE values, rather than obtaining its gain only at a few isolated SNR points.

Fig. 6 reports the NMSE from $0.2R_R$ to $1.2R_R$ at 20 dB. The proposed method maintains the lowest NMSE among the temporally refined methods throughout the tested range, while its error increases gradually with distance. This behavior follows the near-field observation model in Sec. II: increasing range progressively weakens the spherical-wave curvature that encodes position, thereby reducing geometric sensitivity. Accordingly, no abrupt performance change is observed at the conventional Rayleigh boundary R_R.

C. Performance under Representative Operating Scenarios

We next examine the proposed framework under three representative variations: increasing motion intensity, changing spatial and temporal observation support, and propagation-model mismatch. These experiments respectively probe the effects of trajectory-model bias, available geometric information, and unmodeled coherent residuals induced by an NLoS path.

1) Smooth accelerated motion: We first vary the trajectory complexity through the RMS acceleration

$$a_{\mathrm{RMS}} = \left[\frac{1}{t_{\mathrm{e}} - t_{\mathrm{s}}} \int_{t_{\mathrm{s}}}^{t_{\mathrm{e}}} \left\| \frac{\mathrm{d}^2 \mathbf{p}(t)}{\mathrm{d}t^2} \right\|_2^2 \mathrm{d}t \right]^{1/2}. \tag{45}$$

As shown in Fig. 7, the proposed method achieves the lowest NMSE for $a_{\mathrm{RMS}} \leq 1.4\ \mathrm{m/s^2}$. However, when the motion becomes more complex, the fixed smoothness prior increasingly suppresses genuine trajectory variations, leading to a larger regularization bias. This limitation is also partly associated with the adopted open-uniform knot sequence, which allocates the spline resolution uniformly over time regardless of the local motion intensity. A natural extension is therefore to adopt nonuniform or adaptive knot placement, assigning denser spline support to intervals with stronger motion variations in future work.

2) Spatial and temporal observation support: Figs. 8 and 9 examine how the proposed method benefits from increased spatial and temporal observation support, respectively.

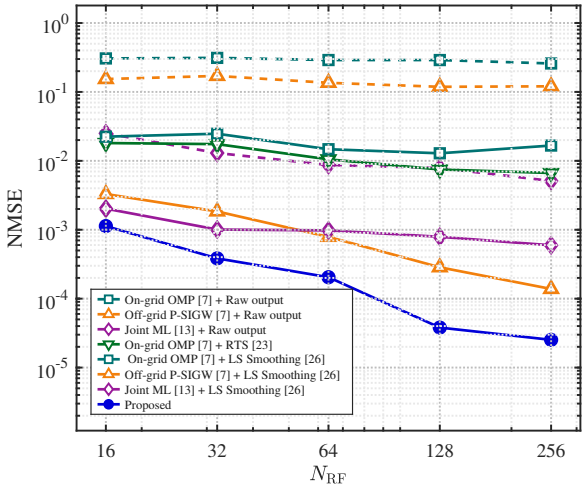

Fig. 8. NMSE versus combiner output dimension.

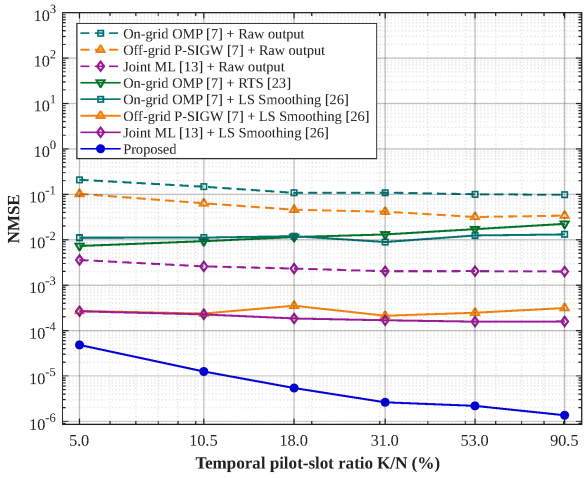

Fig. 9. NMSE versus temporal pilot-slot ratio.

In Fig. 8, increasing N_{RF} consistently improves the proposed method. Since the nested semiunitary combiners preserve the physical aperture, this gain results from retaining more post-combining response dimensions. As characterized by equations (18),(19), only the geometry-sensitive response variations orthogonal to the nuisance direction of α_k contribute efficient trajectory information. Increasing N_{RF} preserves more such components, thereby strengthening the pilot-local information and its mapped constraint on the coefficient vector.

Fig. 9 further varies the temporal pilot support by increasing K. The proposed method maintains the lowest NMSE over the tested range and generally improves as the pilot density increases. Unlike Fig. 8, which strengthens the information supplied by each individual pilot, increasing K introduces more pilot-local contributions into equation (21) and provides denser temporal sampling of the shared spline. According to Proposition 1, these additional mapped contributions improve the coverage and conditioning of the shared coefficient space, thereby reducing the trajectory estimation error.

3) Unmodeled NLoS-path mismatch: We then examine propagation-model mismatch by adding one unmodeled NLoS reflected path to the observations while all methods retain the LoS-based likelihood. The NLoS-to-LoS power ratio is varied from the LoS-only case to -5 dB.

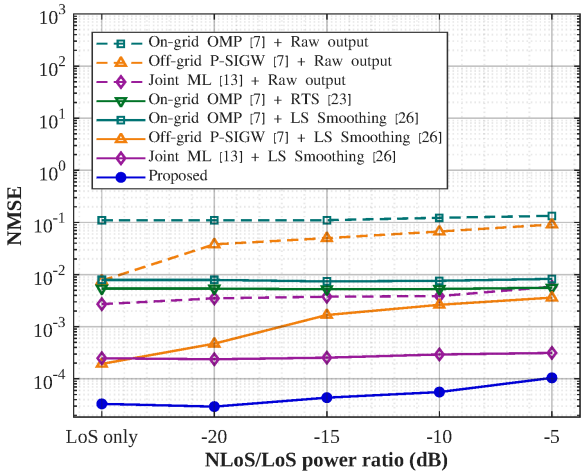

Fig. 10. NMSE versus NLoS-to-LoS power ratio with one unmodeled reflected path.

As shown in Fig. 10, the proposed method maintains the lowest NMSE over all tested mismatch levels. For weak reflections, the shared coefficient vector prevents individual pilot positions from independently drifting to absorb the unmodeled component, thereby suppressing mismatch-induced local deviations. As the reflected power increases, however, its residual component after nuisance projection increasingly overlaps the geometry-sensitive directions characterized by $\mathbf{P}^{\perp}_{\psi,k}\mathbf{Q}_{\boldsymbol{\varpi},k}$. When such components remain coherent across pilot instants, they accumulate in the shared coefficient space and gradually bias $\boldsymbol{\varpi}$.

The proposed method nevertheless retains the lowest NMSE throughout the tested dominant-LoS mismatch range, with gradual degradation as the reflected component strengthens.

D. Algorithmic Properties and Efficiency

We finally examine the numerical and computational behavior of the proposed solver from four aspects, including regularization sensitivity, optimization behavior, structural acceleration of the alternating updates, and the resulting end-to-end accuracy–runtime trade-off.

1) Regularization and trajectory complexity: We first examine how regularization interacts with trajectory complexity. Here, ground-truth and estimated trajectories use the same cubic B-spline family and the same L, so varying L changes trajectory complexity without introducing representation mismatch. Fig. 11 compares three regularization strengths for trajectories with $L = 4$ and $L = 32$ over the tested SNR range. The no-, nominal-, and strong-regularization settings correspond to $(\lambda_1, \lambda_2) = \mathbf{0}$, the nominal values in Table IV, and 10^3 times the nominal values respectively.

For $L = 4$, the default $K = 10$ satisfies the necessary dimension condition $K \geq L$ in Proposition 1, and under the tested pilot geometry the low-dimensional coefficient space is sufficiently constrained by the observations. Accordingly,

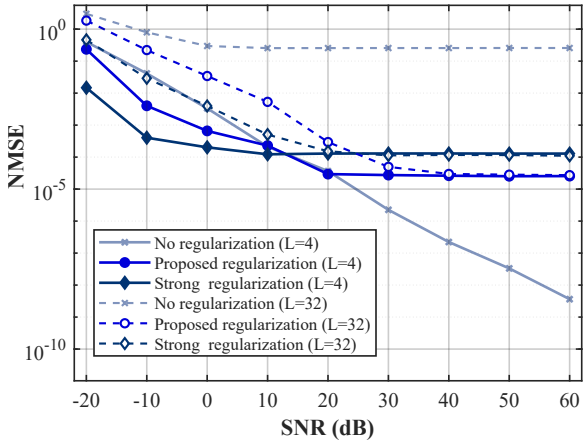

Fig. 11. NMSE versus SNR under different regularization strengths and trajectory complexities.

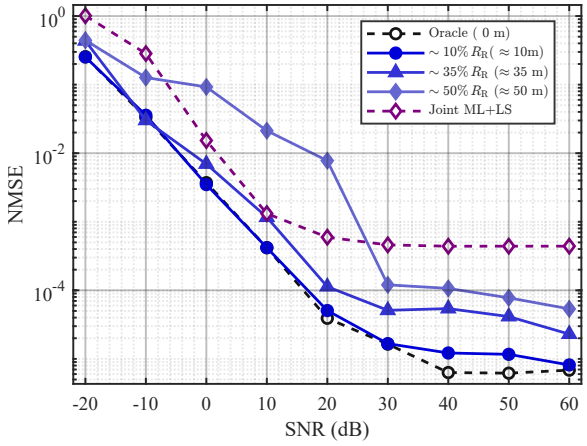

Fig. 12. NMSE versus SNR under different initialization perturbations.

the unregularized estimator continues to improve with SNR without an appreciable error floor, whereas regularization eventually introduces a high-SNR bias floor.

In contrast, $L = 32 > K$ violates the necessary information-dimension condition, making the observation-level coefficient information rank deficient regardless of SNR. The unregularized estimator therefore exhibits severe performance degradation, while regularization supplies curvature along weakly or unsupported coefficient directions and substantially stabilizes the estimation. Excessive regularization, however, introduces additional smoothing bias, as reflected by the higher floor of the strong-regularization setting.

Fig. 11 therefore confirms the distinction in Sec. III-B. Observation-supported directions improve with SNR, whereas regularization mainly stabilizes unsupported directions at the cost of high-SNR bias.

2) Optimization landscape and basin selection: We next examine the sensitivity to initialization and the role of the proposed lifted local–global formulation in navigating the nonconvex landscape.

Fig. 12 evaluates initialization offsets of approximately $0.1R_R$, $0.35R_R$, and $0.5R_R$. As expected, larger perturbations mainly degrade performance when the observations are weak.

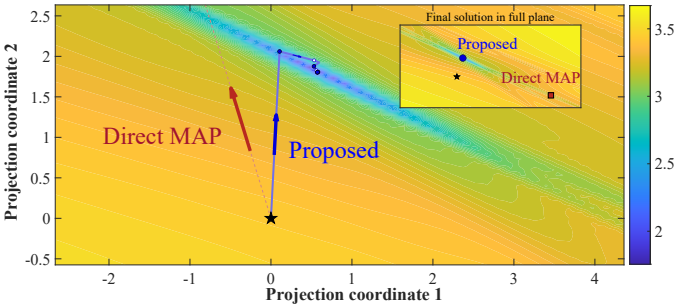

Fig. 13. Solution-spanning slice of $\log_{10} J_{\mathrm{dir}}$ at 20 dB, with the projected optimization paths of the lifted solver and safeguarded Direct MAP.

With increasing SNR, however, even the $0.35R_R$ and $0.5R_R$ initializations recover substantially, while the $0.1R_R$ case approaches the oracle initialization. This indicates that sufficiently informative pilot observations can correct considerable initialization errors, whereas weak observations combined with a distant initialization increase the likelihood of convergence to unfavorable basins.

To further isolate the effect of variable lifting, we compare the proposed solver with safeguarded Direct MAP, which minimizes $J_{\mathrm{dir}}(\boldsymbol{\varpi})$ directly from the same observations and initialization but without the auxiliary variables or penalty continuation. Fig. 13 projects their iterates onto a two-dimensional solution-spanning subspace, with the background showing $\log_{10} J_{\mathrm{dir}}$. Although both methods start from the same point, their optimization paths rapidly diverge. Direct MAP follows the coupled objective directly and is attracted to a different local basin, whereas the lifted solver follows the local–global continuation toward the lower-objective valley and reaches a more accurate terminal solution. This behavior illustrates how relaxing local–global consistency through the auxiliary variables at early continuation stages allows the pilot-local geometry to adjust before being progressively reconciled with the shared trajectory, thereby steering the solver toward a different basin of the nonconvex objective.

3) Structured and parallel implementation efficiency: We next validate the two structural accelerations developed in Sec. IV, namely, parallel local updates and the compact-banded global solve.

Fig. 14 validates the local parallelization. As K increases from 8 to 20, serial latency grows nearly linearly, whereas the parallel implementation increases slowly. At $K = 20$, runtime decreases from 0.129 s to 0.049 s (2.6×).

Fig. 15 measures the time of each repeated global update after the continuation-stage factorization is reused. Dense, generic-sparse, and compact-banded solvers are comparable for small L, but diverge as L grows; the proposed backend follows the predicted near-linear scaling and becomes substantially cheaper at $L = 512$.

Figs. 14 and 15 experimentally confirm that parallelization suppresses the K-dependent local latency, while fixed-bandwidth banded algebra prevents the global solve from growing quadratically or cubically with L. Their combination forms the computational basis for the end-to-end accuracy–runtime trade-off evaluated next.

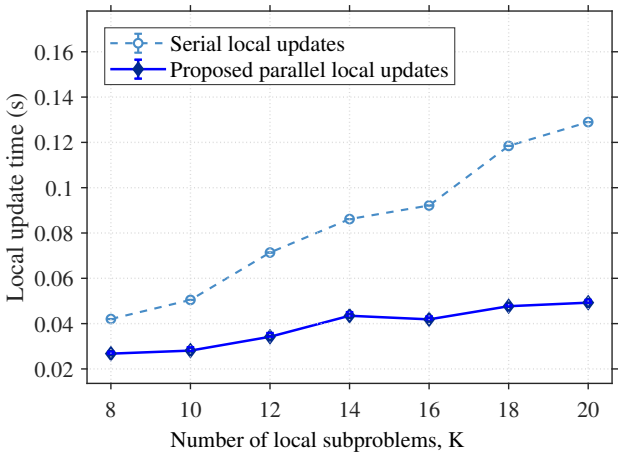

Fig. 14. Local-block time under serial and parallel execution.

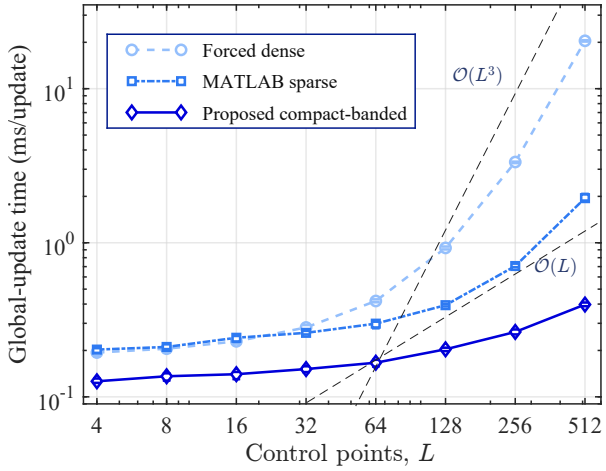

Fig. 15. Time of each repeated global update versus control-point count.

4) End-to-end accuracy–runtime trade-off: The preceding experiments separately examine the numerical behavior and the local–global structural efficiency of the proposed solver. Fig. 16 further compares the complete inference pipelines in terms of end-to-end runtime and NMSE. Since the SNR dependence has already been systematically characterized in Fig. 5, we fix the SNR at the representative value of 10 dB here to isolate the accuracy–runtime variation induced by the baseline search/codebook configurations.

For each snapshot-first baseline, following the polar-domain codebook construction in [7], we vary the angular sampling rate $R_{sample} \in \{1, 2, 4\}$ and the range-sampling parameter β over six tested values, resulting in 18 tested (R_{sample}, β) configurations. For each configuration, 20 Monte Carlo trials are performed, and their mean runtime and mean NMSE form one configuration-level operating point. The empirical covariance ellipse in Fig. 16 is computed in the $(\log_{10}(\text{runtime}), \log_{10}(\text{NMSE}))$ plane from these 18 configuration-level mean points. Therefore, each ellipse characterizes how the average accuracy–runtime operating point of a baseline varies with its codebook/search parameters, rather than the trial-to-trial Monte Carlo variability.

The major-axis direction and extent of each ellipse characterize the dominant joint variation of runtime and NMSE as (R_{sample}, β) changes, while its tilt indicates the prevailing accuracy–runtime trade-off. The minor-axis extent describes the residual configuration-dependent spread around this dominant trend, and the ellipse area summarizes the overall sensitivity of the operating point to the tested codebook settings. Accordingly, a more compact ellipse indicates greater robustness to codebook/search-parameter selection, whereas a broader ellipse indicates stronger configuration dependence.

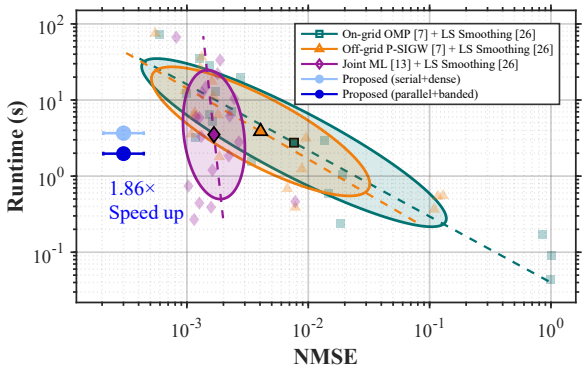

Fig. 16. End-to-end runtime–NMSE operating regions at 10 dB constructed from 18 (R_{sample}, β) configuration-level mean points.

The OMP- and P-SIGW-based ellipses exhibit pronounced negative-slope major axes, showing that increasing the spatial search resolution generally improves accuracy at the cost of additional computation. The extremely fast operating points correspond to very coarse or small codebooks, which substantially reduce the number of candidate evaluations but also lead to sharp NMSE degradation. Hence, these points represent the low-computation, low-accuracy end of the corresponding accuracy–runtime trade-off rather than comparable accuracy at substantially lower cost. In contrast, the Joint-ML ellipse is more vertically oriented, indicating that the tested configurations produce substantial runtime variation with comparatively limited systematic NMSE improvement.

The proposed estimator does not depend on the baseline codebook parameters R_{sample} and β and therefore has no corresponding configuration ellipse. Instead, the serial–dense and parallel–banded implementations are represented by their mean runtime–NMSE operating points together with Monte Carlo error bars, which characterize their trial-to-trial variability. Both proposed operating points lie on the lower-NMSE side of the tested baseline operating regions at comparable runtime. This indicates that, within the tested search range, the accuracy gain of the proposed motion-native inference is not reproduced simply by retuning the snapshot-first codebook resolution.

Finally, replacing the serial–dense backend with the parallel–banded implementation yields a $1.86\times$ runtime reduction with essentially unchanged NMSE. This comparison isolates the implementation-level acceleration of the same proposed estimator and should be distinguished from the accuracy–runtime trade-off across different baseline configurations. Overall, Fig. 16 demonstrates a favorable tested accuracy–runtime operating point without implying a universal

runtime advantage over all possible implementations or search configurations.

VI. Conclusion

In this paper, we developed a motion-native framework for continuous channel tracking in THz near-field XL-MIMO systems by treating the UE trajectory over each tracking window as the primary inference object. A cubic B-spline representation couples sparse pilot observations through one shared trajectory before any pilot-local state is finalized. We characterized the observation-supported identifiability of the shared trajectory and showed that trajectory-relevant information lost during parameter-independent snapshot-first processing cannot, in general, be recovered by subsequent temporal processing. These results clarify when complementary pilot observations can jointly constrain the trajectory and why direct observation-level coupling can outperform estimate-then-track or estimate-then-smooth pipelines.

To solve the resulting nonlinear problem efficiently, we introduced a lifted local–global formulation with parallel pilot-local geometric updates and a structured global spline update. The compact support of the B-spline basis yields a fixed-bandwidth global system with linear-complexity factorization for a fixed spline degree. Simulations demonstrate improved tracking accuracy over representative snapshot-first baselines across the tested SNR, propagation-range, observation-support, smooth-motion, and dominant-path-mismatch settings. The results further characterize regularization sensitivity, optimization behavior, and the end-to-end accuracy–runtime trade-off enabled by the structured implementation.

The present formulation focuses on a smooth-motion, dominant-trackable-path, single-user batch setting. Future work will extend the framework to wideband hybrid-field and multipath propagation, investigate physics-constrained deep unfolding of the structured solver, and develop joint multi-user tracking under practical synchronization, calibration, and interference impairments.

Appendix A
Proofs of the Identifiability Results

Proof: We first establish the general rank characterization (22) in Proposition 1. By construction, each block of $\mathbf{A}_{\mathcal{O}}$ is the pilot-local geometric information weighted by its spline mapping. Therefore, stacking all pilot contributions gives the Gram representation

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} = \sum_{k=1}^{K} \mathbf{S}_k^{\mathsf{T}} \mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} \mathbf{S}_k = \mathbf{A}_{\mathcal{O}}^{\mathsf{T}} \mathbf{A}_{\mathcal{O}}. \tag{46}$$

This representation directly connects observation-supported identifiability to the column space of $\mathbf{A}_{\mathcal{O}}$. In particular, for any nonzero $\delta\boldsymbol{\varpi} \in \mathbb{R}^{2L}$,

$$\delta\boldsymbol{\varpi}^{\mathsf{T}} \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \delta\boldsymbol{\varpi} = \|\mathbf{A}_{\mathcal{O}} \delta\boldsymbol{\varpi}\|_2^2 . \tag{47}$$

Hence, $\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \succ \mathbf{0}$ holds if and only if $\mathbf{A}_{\mathcal{O}}$ has no nontrivial null direction, i.e., $\mathrm{rank}(\mathbf{A}_{\mathcal{O}}) = 2L$. The same representation also yields the necessary information-dimension condition. If $\mathrm{rank}(\mathbf{A}_{\mathcal{O}}) = 2L$, then

$$\begin{aligned} 2L &= \mathrm{rank}(\mathbf{A}_{\mathcal{O}}) \\ &\le \sum_{k=1}^{K} \mathrm{rank}\left((\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}})^{1/2} \mathbf{S}_k \right) \\ &\le \min\left\{ \sum_{k=1}^{K} \mathrm{rank}\left((\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}})^{1/2} \right), \sum_{k=1}^{K} \mathrm{rank}(\mathbf{S}_k) \right\} \\ &= \min\left\{ \sum_{k=1}^{K} \mathrm{rank}\left(\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} \right), 2K \right\} \le 2K. \end{aligned} \tag{48}$$

Since $\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} \in \mathbb{R}^{2\times 2}$ and $\mathrm{rank}(\mathbf{S}_k) = 2$, (48) implies both $\sum_{k=1}^{K} \mathrm{rank}\left(\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}\right) \ge 2L$ and $K \ge L$.

Beyond the preceding condition, we next establish the B-spline sampling condition in (23). The necessary dimension condition $K \ge L$ alone does not guarantee identifiability, because the pilot sampling pattern must also span the complete B-spline coefficient space. From the definition of $\mathbf{A}_{\mathcal{O}}$,

$$\mathbf{A}_{\mathcal{O}} \delta\boldsymbol{\varpi} = \begin{bmatrix} (\mathcal{I}_{\mathbf{p},1}^{\mathrm{obs}})^{1/2} \mathbf{S}_1 \delta\boldsymbol{\varpi} \\ \vdots \\ (\mathcal{I}_{\mathbf{p},K}^{\mathrm{obs}})^{1/2} \mathbf{S}_K \delta\boldsymbol{\varpi} \end{bmatrix}. \tag{49}$$

If the kth pilot has full local identifiability, i.e., $\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}} \succ 0$ for every k, then each $(\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}})^{1/2}$ is nonsingular. Therefore, the information weighting does not change the pilot-wise null directions, and

$$\mathbf{A}_{\mathcal{O}} \delta\boldsymbol{\varpi} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{S}_k \delta\boldsymbol{\varpi} = \mathbf{0}, \ \forall k. \tag{50}$$

Since $\mathbf{S}_k = \mathbf{b}^{\mathsf{T}}(t_{n_k}) \otimes \mathbf{I}_2$, the stacked spline constraints are equivalently represented by $\mathbf{B}_{\mathcal{O}} \otimes \mathbf{I}_2$. Therefore,

$$\mathrm{rank}(\mathbf{A}_{\mathcal{O}}) = \mathrm{rank}(\mathbf{B}_{\mathcal{O}} \otimes \mathbf{I}_2) = 2\,\mathrm{rank}(\mathbf{B}_{\mathcal{O}}). \tag{51}$$

Combining this relation with (22) gives

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} \succ \mathbf{0} \quad \Longleftrightarrow \quad \mathrm{rank}(\mathbf{B}_{\mathcal{O}}) = L,$$

which proves (23). For a degree-q B-spline, the latter full-column-rank condition is guaranteed if the pilot set contains L ordered sites satisfying the Schoenberg–Whitney conditions $B_{i,q}(\widetilde{s}_i) > 0$ [30]. Indeed, the corresponding $L \times L$ row submatrix of $\mathbf{B}_{\mathcal{O}}$ is nonsingular, which implies $\mathrm{rank}(\mathbf{B}_{\mathcal{O}}) = L$. This completes the proof. ∎

Appendix B
Proof of Proposition 2

Proof: Considering the complete snapshot-first processing chain $\widehat{\boldsymbol{\xi}}_k = \mathcal{K}_{\mathrm{s},k}(\mathbf{y}_k)$ in (25), both mappings are parameter independent under the standard regularity conditions. Thus, the data-processing inequality implies that a subsequent parameter-independent statistical mapping cannot increase efficient Fisher information as

$$\mathcal{I}_{\mathbf{p},k}^{\mathrm{snap}} \preceq \mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}, \qquad k = 1, \ldots, K. \tag{52}$$

Since $\mathbf{p}_k = \mathbf{S}_k \boldsymbol{\varpi}$, the chain rule maps the efficient information retained by the kth snapshot into the coefficient space as

$\mathbf{S}_k^{\mathsf{T}}\mathcal{I}_{\mathbf{p},k}^{\mathrm{snap}}\mathbf{S}_k$. Moreover, because the snapshot mappings remain pilot-wise and the pilot observations are conditionally independent, the information retained by the complete snapshot sequence with respect to $\boldsymbol{\varpi}$ is

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}} \triangleq \sum_{k=1}^{K} \mathbf{S}_k^{\mathsf{T}}\mathcal{I}_{\mathbf{p},k}^{\mathrm{snap}}\mathbf{S}_k. \tag{53}$$

Similarly, Proposition 1 gives the raw-observation information

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} = \sum_{k=1}^{K} \mathbf{S}_k^{\mathsf{T}}\mathcal{I}_{\mathbf{p},k}^{\mathrm{obs}}\mathbf{S}_k. \tag{54}$$

Subtracting (53) from (54) and using (26) yields

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}} = \sum_{k=1}^{K} \mathbf{S}_k^{\mathsf{T}}\Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}}\mathbf{S}_k \succeq \mathbf{0}. \tag{55}$$

Hence, information discarded during pilot-local snapshot formation is mapped into the shared coefficient space through the same spline operators that aggregate complementary pilot information in Proposition 1.

We then consider the subsequent temporal processing $\widehat{\boldsymbol{\zeta}} = \mathcal{K}_{\mathrm{p}}(\widehat{\boldsymbol{\xi}})$. Applying the same data-processing principle to this parameter-independent mapping, now with respect to the shared trajectory parameter $\boldsymbol{\varpi}$, gives

$$\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{post}} \preceq \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}}. \tag{56}$$

Therefore, the additional trajectory-information loss introduced by temporal postprocessing is

$$\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} \triangleq \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{post}} \succeq \mathbf{0}. \tag{57}$$

Finally, using $\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} \triangleq \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{post}}$, we decompose

$$\begin{aligned}\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} &= \left(\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{obs}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}}\right) + \left(\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{snap}} - \mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{post}}\right) \\ &= \sum_{k=1}^{K} \mathbf{S}_k^{\mathsf{T}}\Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}}\mathbf{S}_k + \Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} \succeq \mathbf{0},\end{aligned} \tag{58}$$

which proves (27). It remains to establish the zero-loss condition in (28).

If $\Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}} = \mathbf{0}$ for every k and $\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} = \mathbf{0}$, then (58) immediately gives $\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} = \mathbf{0}$. Conversely, suppose that $\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathrm{tot}} = \mathbf{0}$. All terms on the right-hand side of (58) are positive semidefinite, and therefore none can cancel another. It follows that

$$\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} = \mathbf{0}, \qquad \mathbf{S}_k^{\mathsf{T}}\Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}}\mathbf{S}_k = \mathbf{0}, \quad \forall k. \tag{59}$$

For the adopted B-spline representation, $\mathbf{S}_k = \mathbf{b}^{\mathsf{T}}(t_{n_k}) \otimes \mathbf{I}_2$. Since the B-spline basis forms a partition of unity, $\mathbf{b}(t_{n_k}) \neq \mathbf{0}$ at every valid pilot instant, and hence $\mathrm{rank}(\mathbf{S}_k) = 2$. Thus, $\mathbf{S}_k$ has full row rank, yielding $\Delta\mathcal{I}_{\mathbf{p},k}^{\mathcal{K}_{\mathrm{s}}} = \mathbf{0}$ for all k. Together with $\Delta\mathcal{I}_{\boldsymbol{\varpi}}^{\mathcal{K}_{\mathrm{p}}} = \mathbf{0}$, this proves (28). ∎

APPENDIX C
PROOFS OF THE B-SPLINE STRUCTURAL PROPERTIES

Proof of Lemma 1: For $q = 0$, the definition of the degree-zero basis gives $\mathrm{supp}(B_{\ell,0}) \subseteq [\tau_\ell, \tau_{\ell+1}]$. Assume that

$$\mathrm{supp}(B_{\ell,q-1}) \subseteq [\tau_\ell, \tau_{\ell+q}].$$

By the Cox–de Boor recursion in (5), $B_{\ell,q}$ is a weighted sum of $B_{\ell,q-1}$ and $B_{\ell+1,q-1}$. Since the scalar weights do not enlarge their supports,

$$\begin{aligned}\mathrm{supp}(B_{\ell,q}) &\subseteq \mathrm{supp}(B_{\ell,q-1}) \cup \mathrm{supp}(B_{\ell+1,q-1}) \\ &\subseteq [\tau_\ell, \tau_{\ell+q+1}].\end{aligned}$$

The support bound therefore follows by induction. If $s \in [\tau_j, \tau_{j+1})$, only the consecutive indices $\ell \in \{j-q, \ldots, j\}$ can be active. Hence, at most $q+1$ basis functions are nonzero at any $s \in [0,1]$, including the right endpoint under the adopted clamped-knot convention. ∎

Proof of Lemma 2: By Lemma 1, at most $q+1$ consecutive B-spline basis functions are active at any pilot instant. Consequently, $\mathbf{S}_k$ contains at most $q+1$ consecutive active control-point blocks, and $[\mathbf{S}_k^{\mathsf{T}}\mathbf{S}_k]_{\ell,r} = \mathbf{0}_{2\times 2}$ whenever $|\ell - r| > q$.

Moreover, $\mathbf{D}_1$ and $\mathbf{D}_2$ are the first- and second-order difference operators, so $\mathbf{D}_1^{\mathsf{T}}\mathbf{D}_1$ and $\mathbf{D}_2^{\mathsf{T}}\mathbf{D}_2$ have block half-bandwidths at most one and two, respectively. Hence, the regularization matrix $\mathbf{R}$ has block half-bandwidth at most two. Since $\mathbf{C}_{\boldsymbol{\varpi}} = \mathbf{R} + \eta\sum_{k=1}^{K}\mathbf{S}_k^{\mathsf{T}}\mathbf{S}_k$, it follows that

$$[\mathbf{C}_{\boldsymbol{\varpi}}]_{\ell,r} = \mathbf{0}_{2\times 2}, \qquad |\ell - r| > \max\{q, 2\}.$$

Therefore, $\mathbf{C}_{\boldsymbol{\varpi}}$ is symmetric and block banded with block half-bandwidth at most $b_C = \max\{q, 2\}$, independently of L. ∎

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